Fourier Transforms in Circuit Analysis

Directed Reading Program in Mathematics
Seo Bin Han
Advised by: Connor Fitzgerald
University of California, Davis, CA 95616

Introduction of the Fourier Transform

The Fourier Transform converts a time domain function into its frequency domain representation:

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt$$

One of the most useful properties of the Fourier Transform is how it converts differentiation to multiplication, which simplifies the analysis of differential equations:

$$\mathcal{F}(f'(t)) = \int f'(t)e^{-i\omega t}dt$$

$$\boxed{\int f'(t)e^{-i\omega t}dt = \left[f(t)e^{-i\omega t}\right]_{-\infty}^{\infty} - \int f(t)(-i\omega)e^{-i\omega t}dt}$$

$u = e^{-i\omega t}$ $\quad dv = f'(t)dt$

$du = -i\omega e^{-i\omega t}dt$ $\quad v = f(t)$

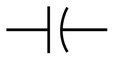
$$= i\omega \int f(t)e^{-i\omega t}dt$$

$$= i\omega F(\omega)$$

$$\mathcal{F}(f'(t)) = i\omega \hat{f}(\omega)$$

By considering the derivative of a Fourier Transform, then performing integration by parts, we were able to find that taking the Fourier Transform of a derivative simply corresponds to multiplying the Fourier Transform of the function by iω.

RLC Time Response

Resistor	Inductor	Capacitor
Opposes the flow of current, causing a voltage drop that is directly proportional to the current. We call the proportionality constant the resistance, R. $V = IR.$	Resists changes in current by generating a voltage proportional to the rate of change of current. We call the proportionality constant the inductance, L. $V = L\frac{dI}{dt}.$	Stores electrical charge and relates voltage to the amount of charge it holds. The voltage across a capacitor is related to charge and capacitance. $V = \frac{Q}{C}.$
		

RLC Frequency Response

The frequency response of circuit elements is characterized by their impedance and the general relationship is as follows:

$$\hat{V}(\omega) = Z(\omega)\hat{I}(\omega)$$

For a resistor (R),
We denote Ohm's Law in the time domain:
Using the FT described by section 1,
We find the impedance as:

$$V(t) = RI(t)$$
$$\hat{V}(\omega) = R\hat{I}(\omega)$$
$$Z(\omega) = R$$

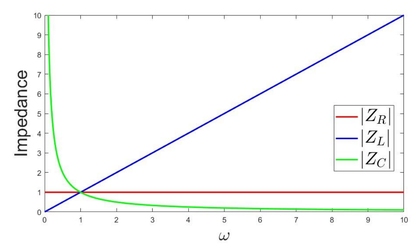
For an inductor (I),

$$V(t) = L\frac{dI}{dt}$$
$$\hat{V}(\omega) = L\mathcal{F}\left(\frac{dI}{dt}\right) = L(i\omega\hat{I}(\omega)) = (i\omega L)\hat{I}(\omega)$$
$$Z(\omega) = i\omega L$$

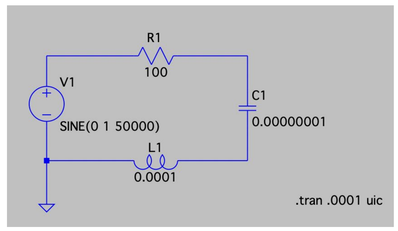
For a capacitor (C),

$$\frac{dV}{dt} = \frac{I}{C}$$
$$\mathcal{F}\left(\frac{dV}{dt}\right) = \frac{1}{C}\hat{I}(\omega)$$
$$i\omega V(\omega) = \frac{1}{C}I(\omega)$$
$$Z(\omega) = \frac{1}{i\omega C} = \frac{-i}{\omega C}$$

The corresponding graphs can be represented as follows:



RLC Circuit via Kirchhoff's Voltage Law



A series RLC circuit consists of a resistor (R), inductor (L), and capacitor (C). Applying Kirchhoff's Voltage Law (KVL) to the circuit gives:

$$V_{in} = V_R + V_L + V_C = IR + L\frac{dI}{dt} + \frac{1}{C}\int Idt.$$

Loop Law Analysis

Following the equation we derived from KVL, we can take the derivative of both sides to get a non-homogeneous 2nd-order constant coefficient ODE.

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{1}{C}I = \frac{dV_{\text{in}}}{dt}$$

By applying the Fourier Transform, we express voltages and currents in the frequency domain using phasor notation. The total impedance of the RLC circuit in series is:

$$Z_{\text{tot}}(\omega) = Z_R + Z_L + Z_C = R + i\left(\omega L - \frac{1}{\omega C}\right)$$

Thus, Ohm's Law in the frequency domain gives:

$$V_{\text{in}}(\omega) = Z_{\text{tot}}(\omega)I(\omega)$$

Solving for I(ω):

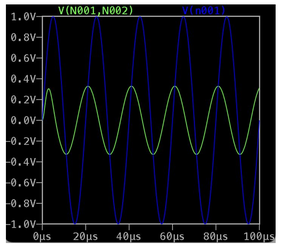
$$I(\omega) = \frac{V_{\text{in}}(\omega)}{R + i\left(\omega L - \frac{1}{\omega C}\right)}$$

The voltage across the resistor is therefore:

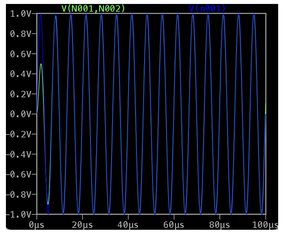
$$V_R(\omega) = RI(\omega) = \frac{V_{\text{in}}(\omega)R}{R + i\left(\omega L - \frac{1}{\omega C}\right)}$$

$$\left|\frac{V_R}{V_{\text{in}}}\right|^2 = \frac{R^2}{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \frac{\left(\frac{R\omega}{L}\right)^2}{\left(\frac{R\omega}{L}\right)^2 + \left(\omega^2 - \frac{1}{LC}\right)^2}$$

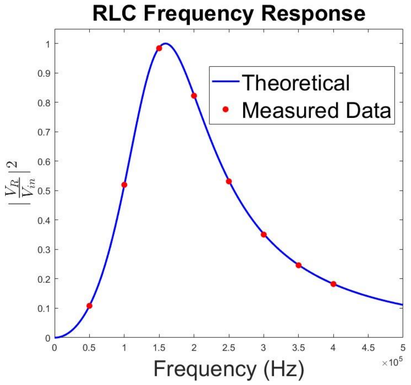
Data Analysis in LTSpice



Frequency = 50,000 Hz



Frequency = 150,000 Hz



The circuit exhibited a resonance peak around 150,000 Hz, where the voltage ratio reaches its maximum. At frequencies far from resonance, the response diminishes, as expected from the impedance characteristics of an RLC circuit.